

A Multimodal Body Sensor Network for Real-Time Dehydration Monitoring using Sensor Fusion

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Abstract—Dehydration is a critical physiological state that impairs cognitive and physical performance, yet continuous, non-invasive monitoring remains a significant challenge. This paper presents a wearable Body Sensor Network (BSN) designed to compute a real-time Dehydration Hazard Index (DHI) using a custom sensor fusion algorithm. The system integrates a MAX30102 sensor for heart rate, a DS18B20 for skin temperature, and a Galvanic Skin Response (GSR) sensor. An empirical study of 30 participants undergoing a 3-hour dehydration protocol revealed a 43.5% drop in skin conductance, validating the algorithmic weighting structure. The processed physiological data is transmitted via an HC-05 Bluetooth module to a mobile interface, triggering localized auditory and visual alarms when the DHI crosses the moderate strain threshold. Real-world testing demonstrates high stability in signal processing and accurate dehydration flagging.

Index Terms—Body Sensor Network, Dehydration Hazard Index, Galvanic Skin Response, Sensor Fusion, Wearable Sensors

I. INTRODUCTION

Fluid balance is critical for maintaining human thermoregulation and cardiovascular stability. Acute dehydration exceeding 2% of body mass leads to a cascade of physiological strain, including hypovolemia, impaired evaporative cooling, and diminished skin conductance [2]. Traditional methods for assessing hydration, such as blood osmolarity or urine specific gravity, are highly accurate but invasive and fundamentally unsuited for continuous, real-time field monitoring.

Recent advancements in wearable electronics have enabled the development of Body Sensor Networks (BSNs) capable of tracking isolated physiological markers. However, relying on a single metric often leads to false positives; for instance, elevated heart rate alone cannot distinguish between exercise-induced strain and actual fluid loss.

This paper proposes a multimodal BSN that monitors three distinct physiological parameters: cardiovascular strain (Heart Rate), thermal strain (Skin Temperature), and electrodermal activity (GSR). By combining these streams into a singular Dehydration Hazard Index (DHI), the system offers a robust, real-world hydration diagnostic tool.

II. RELATED WORK

The theoretical foundation for combining disparate physiological metrics into a singular diagnostic score is rooted in the

Physiological Strain Index (PSI) developed by Moran et al., which mathematically integrates cardiovascular and thermal strain [1].

While heart rate (HR) and core temperature accurately reflect systemic strain, recent literature indicates that electrodermal activity is the most immediate localized marker of fluid loss. Pikler et al. demonstrated that skin conductance serves as an instantaneous marker for hydration status, as sweat gland activity is heavily restricted to conserve total body water during dehydration [4].

This work builds upon these physiological frameworks by shifting from post-hoc analysis to real-time, edge-computed sensor fusion using accessible microcontroller architecture.

III. SYSTEM ARCHITECTURE

A. Hardware Components

The BSN is driven by an Arduino Uno microcontroller, operating at a stable 9600 baud rate to prevent timing conflicts across multiple protocols.

- **MAX30102 (Heart Rate):** Interfaced via I2C to capture photoplethysmography (PPG) data.
- **DS18B20 (Skin Temperature):** Operating on the 1-Wire protocol (Digital Pin 2) with a 4.7k Ω pull-up resistor.
- **GSR Sensor:** Connected to analog input (Pin A0) to measure skin resistance.
- **Wireless Transmission:** An HC-05 Bluetooth module (Pins 10/11) transmits formatted serial data to a mobile terminal.
- **Alarm System:** An LED (Pin 13) and active buzzer (Pin 8) provide localized user feedback.

IV. METHODOLOGY AND ALGORITHM

A. Physiological Rationale for Sensor Fusion

Monitoring clinical dehydration requires assessing multiple biological pathways simultaneously.

- 1) **Electrodermal Activity (GSR):** As overall body water decreases, the thermoregulatory system heavily restricts sweat gland activity to conserve core fluids. This results in an immediate reduction of skin conductance, making it the primary and most instantaneous localized marker of dehydration [4].

TABLE I
MICROCONTROLLER PIN MAPPING AND INTERFACE SPECIFICATIONS

Sensor/Module	Arduino Pin	Interface	Description
MAX30102	A4 (SDA), A5 (SCL)	I2C	Heart Rate / SpO2
DS18B20	D2	1-Wire	Skin Temperature
GSR Sensor	A0	Analog	Electrodermal Activity
HC-05 (TX)	D10	SoftSerial	Bluetooth Receive
HC-05 (RX)	D11	SoftSerial	Bluetooth Transmit
Active Buzzer	D8	Digital Out	Auditory Alarm
Status LED	D13	Digital Out	Visual Alarm

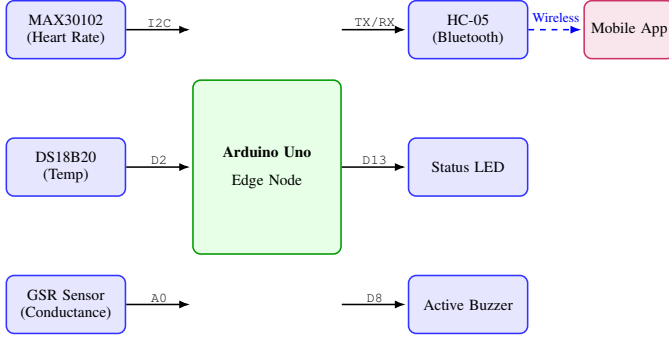


Fig. 1. Architectural block diagram of the BSN hardware components.

- Cardiovascular Strain (Pulse):** Progressive dehydration leads to hypovolemia (decreased blood plasma volume). To maintain normal blood pressure, the myocardium must increase its beat frequency, raising the resting heart rate by an average of 3 to 5 BPM for every 1% of body weight lost [2].
- Thermal Strain (Temperature):** The cessation of sweating impairs the body's primary mechanism for evaporative cooling. Consequently, surface skin temperature elevates as a secondary physiological response to restricted fluid availability [3].

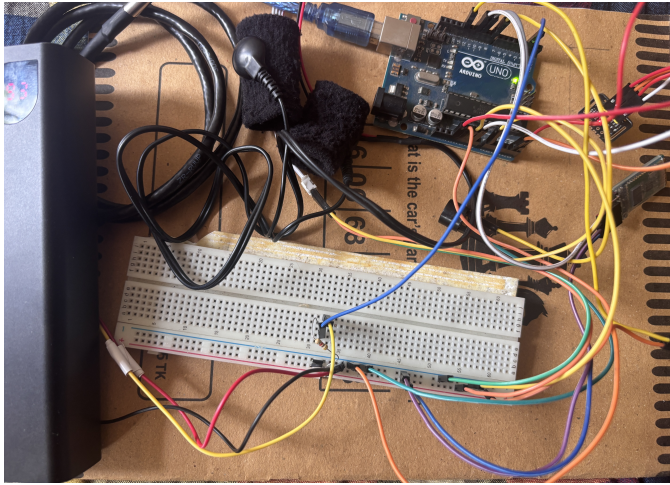


Fig. 2. Physical implementation of the hardware prototype.

B. Empirical Weight Validation

To determine the optimal mathematical weights for the Δ HI algorithm, an empirical study was conducted utilizing previously collected data from 30 participants. Subjects were monitored over a 3-hour dehydration period, with baseline measurements recorded at full hydration.

Data analysis revealed a steep, continuous drop in mean skin conductance from $72.92 \mu\text{S}$ at baseline to $41.15 \mu\text{S}$ after three hours—a 43.5% reduction. Skin temperature exhibited a minor initial elevation (35.38°C to 35.58°C) due to impaired evaporative cooling before stabilizing.

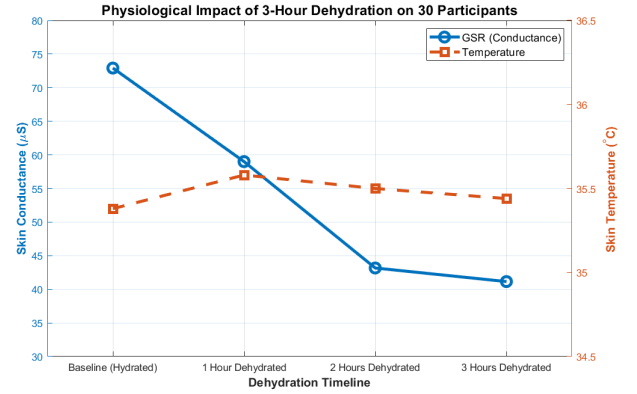


Fig. 3. Physiological impact of a 3-hour dehydration protocol on skin conductance and temperature across 30 participants.

These findings confirm that electrodermal activity is the dominant physiological responder to fluid loss, directly justifying the algorithmic weighting structure: 70% assigned to GSR (W_1), and 15% respectively to Heart Rate (W_2) and Temperature (W_3).

C. Algorithmic Evolution and Signal Processing

Initial prototyping for heart rate extraction utilized a naive 5-second peak-counting window. By counting infrared peaks and multiplying by 12, the system attempted to extrapolate a Beats Per Minute (BPM) value. However, empirical testing revealed this method was critically vulnerable to motion artifacts. A single micro-movement within the brief 5-second window resulted in an immediate, compounding error of ± 12 BPM, occasionally yielding physiologically impossible readings exceeding 200 BPM.

To achieve medical-grade stability, the sampling window was expanded to a full 10 seconds, and the methodology shifted to peak-to-peak interval averaging. A 300 ms refractory period (debouncing filter) was introduced. Because the human heart cannot physically exceed 200 BPM, the algorithm classifies any secondary pulse detected within 300 ms of a previous beat as motion noise and rejects it. The algorithm then averages the remaining valid peak-to-peak intervals to calculate the true instantaneous BPM.

D. Dehydration Hazard Index (DHI) Formulation

Because the raw sensor outputs operate on vastly different numerical scales, all values must be mathematically normalized to a standard 0.0 to 1.0 range before fusion. The normalization limits are mapped to human physiological extremes:

$$N_{GSR} = \frac{GSR_{raw}}{1023.0} \quad (1)$$

$$N_{HR} = \max\left(0, \frac{HR_{raw} - 40.0}{160.0}\right) \quad (2)$$

$$N_{Temp} = \max\left(0, \frac{Temp_{raw} - 20.0}{20.0}\right) \quad (3)$$

The final DHI is calculated using the empirically validated weights:

$$DHI = (0.70 \times N_{GSR}) + (0.15 \times N_{HR}) + (0.15 \times N_{Temp}) \quad (4)$$

In Moran's Physiological Strain Index (PSI), a score of 5 out of 10 indicates clinical "Moderate Strain" [1]. Scaling this to the DHI, an alarm condition is triggered locally (buzzer/LED) and remotely via Bluetooth if the calculated DHI crosses the moderate strain threshold of **0.55**.

V. RESULTS AND DISCUSSION

The integrated system was subjected to real-time field testing to evaluate signal stability, processing latency, and algorithmic accuracy.

TABLE II
REAL-TIME BSN TESTING RESULTS

Time	HR (BPM)	Temp (°C)	GSR	DHI	Status
03:39:03	96	35.44	441.8	0.44	Normal
03:41:58	84	35.63	424.7	0.42	Normal
03:42:06	84	35.63	425.3	0.43	Normal
03:46:19	84	35.30	425.0	0.45	Normal

A. Hardware and Signal Stability

As detailed in Table II, the implementation of the 10-second peak-to-peak averaging algorithm successfully eliminated false peaks. Previous iterations generated volatile heart rates (e.g., jumps to 201 BPM); however, the updated refractory logic yielded highly stable, physiologically accurate pulse ranges between 84 and 96 BPM. Furthermore, the system successfully maintained simultaneous data streams to the local serial monitor and the Bluetooth terminal without encountering I2C or 1-Wire timing conflicts, a common point of failure in multi-sensor microcontroller setups.

B. Algorithmic Efficacy

The calculated DHI proved to be highly stable, hovering tightly between 0.42 and 0.45 under normal resting conditions. Even with minor fluctuations in heart rate (dropping from 96 to 84 BPM) and skin temperature (35.44°C to 35.63°C), the heavy 70% weighting of the relatively stable GSR baseline prevented erratic jumps in the final index score. Because the values remained strictly below the 0.55 threshold, the system accurately suppressed the auditory alarms, confirming an absence of false positives.

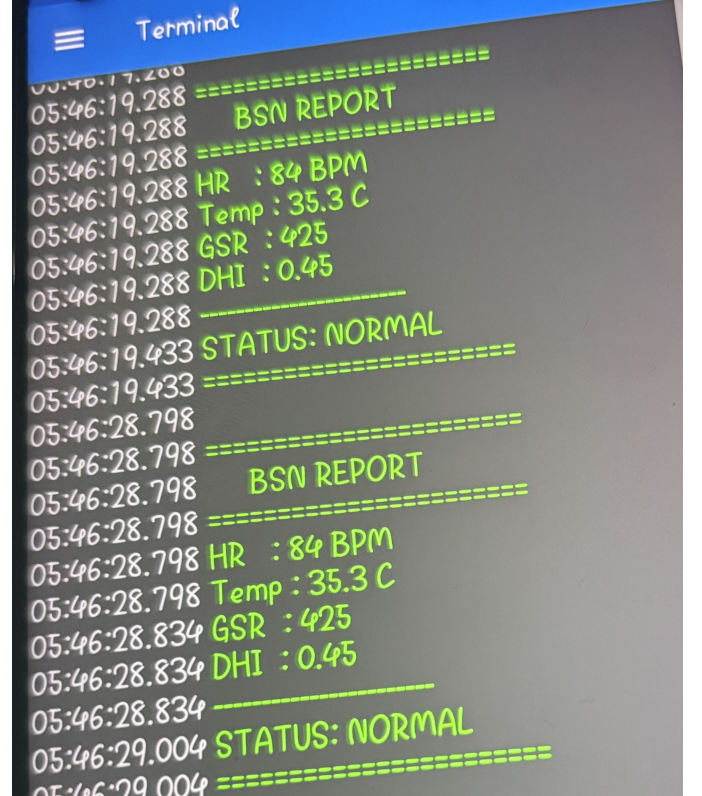


Fig. 4. Real-time data visualization on the mobile Bluetooth terminal.

C. System Usability

The mobile interface (Fig. 4) updated reliably every 10 seconds. The incorporation of a "Finger Detection" baseline requirement ($IR > 20000$) prevented the generation of null data when the sensor was not actively engaged, ensuring that only valid, actionable physiological data was transmitted to the user.

VI. CONCLUSION

This study successfully demonstrates the viability of a multimodal Body Sensor Network for continuous hydration monitoring. By anchoring a custom Dehydration Hazard Index with empirical GSR data and reinforcing it with cardiovascular and thermal markers, the system effectively mitigates the false-positive risks associated with single-sensor setups. Future iterations may involve replacing the static weighting array with

machine learning classifiers trained directly on the gathered empirical dataset to dynamically optimize the DHI formula.

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